Tactix: repaired-source validation and a lost WV100 solve

Tactix Development Project

October 3, 2026

Source `375b621b` passes B28+C5, Rank 1–50 and WV2–89, but **WV100 is NOT ACCEPTED**: rank 96, `snw_16_9_CCSpre`, loses a retained SAT solve at the unchanged 1,200-second deadline. Whole-goal acceptance remains pending. The Substitute watch-restoration repair is unchanged; this checkpoint adds validation evidence, not solver behavior.

Terminal regression evidence

Set	Solver seconds	Checked	Checker TO	Solver TO
B28+C5	956.934	32	1	0
Rank 1–50	7,580.783	41	9	0
WV100	1,743.200	97	2	1 lost
WV2–89	4,092.871	77	2	10 retained
Total	—	247	14	11

All 272 rows are present; 122 SAT models and 125 UNSAT proofs are independently accepted, with no checker rejection. Solver sums exclude timeouts: WV100's smaller sum excludes the lost row and is not an improvement. Fourteen checker timeouts remain censored, not verified. Ten WV2 solver timeouts remain accepted checkpoint exceptions; the additional WV100 timeout rejects its gate.

Arrays preserve full-width submission without throttle/node constraints, 12 logical CPUs, 32,768 MiB and the original solver caps. B28/Rank retain aggregate ceilings 999.87615/8,521.52175 seconds. Each complete proof pipeline has one 1,800-second deadline. Comparable baseline time t allows candidate completion by $t + \max(0.30t, 20\,\mathrm{s})$. All 125 completed comparisons satisfy this bound. Thirteen censored comparisons remain within declared bounds; one already-censored baseline makes its comparison inconclusive. No extended checking campaign was used to claim verification.

The Release solver SHA256 is the concatenation of these two lines:

`b766be0145f7dbc9fd50c03db45ee7e2e`
`3855933faaa84c30865064acd6337ed`

The technical companion binds the baseline, both PIN.json files, manifests, scores, all raw row hashes and terminal scheduler metadata.

Individual instances and the decision-relevant loss

All 261 completed baseline/candidate pairs retain equal normalized printed work counters and certificate hashes, also equal to foundation source `12e3c4f0`. SAT model bytes are not compared in this analysis; candidate models are independently checked. The lost timeout is excluded from

completed correspondence. These facts support recorded default behavior on completed cases, not universal internal-state equivalence.

Completed increases include Rank 8, 79.089 to 98.941 seconds; WV2 rank 245, 29.058 to 47.541; and WV2 rank 244, 23.346 to 39.780. Their page-fault counts increase while recorded solver work stays equal. This suggests execution-cost variation but does not establish page faults as the cause; the rows still pass their actual gates.

The lost WV100 row timed out at 1,200.005 seconds, versus predecessor `f6f01982` SAT in 1,008.042 and baseline SAT in 958.118. All 32 selected inprocessing work rows agree, including 35 phases and 171 Vivify rounds. Inprocessing time rises from 246.544 to 293.898 seconds, Vivify from 141.963 to 181.880. Current final search/certificate and hardware counters are incomplete. Full trajectory equality and CPU/page-fault attribution remain unproved.

SLURM 1747087 runs one sequential predecessor/current pair in the same 12-CPU/32-GiB allocation, same immutable input/arguments, unchanged caps and independent oracles. The predecessor is independently model-verified SAT in 1,046.445 seconds, retaining complete work counters and certificate hash. The candidate remains running at this checkpoint. Order, cache and occupancy variation limit one pair; it cannot replace the original failed full gate.

Feedback overhead and remaining acceptance

Paired feedback job 1746812 completes with exit 0. Its 128 runs cover four fixed synthetics and both policies, one warm-up plus seven measured pairs per group, with alternating arm order. Deterministic signatures agree; 96 SAT models are checked per run, two distinct independently checked certificates cover 32 UNSAT runs, and 64 enabled feedback streams pass strict inspection. Additional median time is 17.597–63.084 milliseconds; enabled/disabled median ratios range from 2.630583 to 8.010679. These are complete frontend costs, including provenance hashing, snapshots, serialization and anonymous-file writes on short inputs. They do not isolate callbacks or establish large-instance overhead. No scalar reward or learning is implemented.

Full Debug 1746924 remains live with 211 passing names and the long scripted-default test outstanding. Exact 212-test JUnit, terminal exits and 984 pre/post bindings remain required; all 984 currently verify. Claude CLI 3818 is live; both fresh fuzz legs are running and targeted jobs queued. No private campaign files were inspected: parent access remains metadata and terminal deliveries. The 100,000-fresh-seed requirement per policy, corpus, effective coverage, purity/replay and negative-oracle acceptance remain pending.

Next: collect the matched candidate/postflight, determine evidence-based WV100 recovery without relaxing the contract, complete Debug and owner acceptance, then run the required recovery gate and whole-objective audit. Preserve all 795 active source bindings, immutable evidence and unrelated job 1745040. No cleanup or push occurred. A metadata-only exporter mistake was corrected before final validation; its aborted unpublished copy is preserved.

See the technical report and its hashed companion for exact scores, correspondence, measurement and interim audit. This report is a checkpoint, not milestone acceptance.